

Numerical

1. Concentration Conversion (ppm to mg/m³)

Problem

The concentration of **carbon monoxide (CO)** in air is **50 ppm** at **25°C** and **1 atm pressure**.
Convert this concentration into **mg/m³**.

(Molecular weight of CO = 28)

Solution

Formula:

$$mg/m^3 = \frac{ppm \times \text{Molecular weight}}{24.45}$$

Why is it used?

At standard conditions (**25°C** and **1 atm**):

- 1 mole of any gas occupies **24.45 liters**

24.45 is valid only at 25°C and 1 atm

$$\begin{aligned} mg/m^3 &= \frac{50 \times 28}{24.45} \\ mg/m^3 &= 57.26 \end{aligned}$$

Answer

CO concentration = **57.26 mg/m³**

2. Particulate Matter Concentration

Problem

A high-volume air sampler collects **0.35 g of dust** in **8 hours**.
The sampler flow rate is **1.5 m³/min**.

Calculate the **concentration of particulate matter (µg/m³)**.

Solution

Total air volume sampled:

$$\text{Flow rate} = 1.5 \text{ m}^3/\text{min}$$

Time:

$$\begin{aligned} 8 \text{ hours} &= 480 \text{ minutes} \\ \text{Total volume} &= 1.5 \times 480 = 720 \text{ m}^3 \end{aligned}$$

Dust collected:

$$0.35 \text{ g} = 350000 \mu\text{g}$$

Concentration:

$$PM = \frac{Mass}{Volume}$$

$$PM = \frac{350000}{720}$$

$$PM = 486.11 \mu\text{g}/\text{m}^3$$

3. Emission Rate Calculation

Problem

An industrial stack emits **SO₂ gas** at a concentration of **500 mg/m³**.
The exhaust gas flow rate is **250 m³/min**.

Calculate the **SO₂ emission rate in kg/hr**.

Solution

Mass emission per minute:

$$Emission = 500 \times 250$$

$$= 125000 \text{ mg}/\text{min}$$

Convert to kg/min:

$$125000 \text{ mg} = 0.125 \text{ kg}$$

Per hour:

$$0.125 \times 60$$

$$= 7.5 \text{ kg}/\text{hr}$$

Answer

SO₂ emission rate = **7.5 kg/hr**

. Cyclone Collector Efficiency

Problem

A cyclone separator receives air containing **3 g/m³ of dust**.
The outlet air contains **0.9 g/m³ of dust**.

Calculate the **collection efficiency of the cyclone collector**.

Solution

Efficiency formula:

$$\eta = \frac{C_{in} - C_{out}}{C_{in}} \times 100$$

Where:

C_{in} = inlet concentration

C_{out} = outlet concentration

$$\eta = \frac{3 - 0.9}{3} \times 100$$

$$\eta = \frac{2.1}{3} \times 100$$

$$\eta = 70\%$$

Answer

Cyclone efficiency = **70%**

5. Electrostatic Precipitator Efficiency

Problem

An electrostatic precipitator treats **10000 m³/min** of flue gas.

Dust concentration entering the ESP is **6 g/m³** and leaving concentration is **0.15 g/m³**.

Calculate:

1. **Efficiency of ESP**
2. **Dust removed per minute**

Step 1: Efficiency

$$\eta = \frac{6 - 0.15}{6} \times 100$$

$$\eta = \frac{5.85}{6} \times 100$$

$$\eta = 97.5\%$$

Step 2: Dust Removed

Dust removed per m³:

$$6 - 0.15 = 5.85 \text{ g/m}^3$$

Total dust removed per minute:

$$\begin{aligned} & 5.85 \times 10000 \\ & = 58500 \text{ g/min} \end{aligned}$$

Convert to kg:

$$58500 = 58.5 \text{ kg/min}$$

Answers

ESP efficiency = **97.5%**

Dust removed = **58.5 kg/min**

Air Quality Index (AQI) Numericals

AQI is calculated using the formula:

$$I = \frac{(I_{HI} - I_{LO})}{(B_{HI} - B_{LO})} (C - B_{LO}) + I_{LO}$$

Where:

- C = pollutant concentration
- B_{HI}, B_{LO} = breakpoint concentrations
- I_{HI}, I_{LO} = AQI values corresponding to breakpoints

AQI Breakpoints for PM2.5 (India)

AQI	PM2.5 ($\mu\text{g}/\text{m}^3$)
0–50	0–30
51–100	31–60
101–200	61–90
201–300	91–120
301–400	121–250
401–500	>250

Air Quality Index (AQI) Classification Table (India)			
AQI Range	Category	Colour Code	Health Impact
0 – 50	Good	Green	Minimal or no impact
51 – 100	Satisfactory	Light Green	Minor breathing discomfort to sensitive people
101 – 200	Moderate	Yellow	Breathing discomfort to people with lung/heart disease, children, elderly
201 – 300	Poor	Orange	Breathing discomfort to most people on prolonged exposure
301 – 400	Very Poor	Red	Respiratory illness on prolonged exposure
401 – 500	Severe	Dark Red / Maroon	Serious health effects; affects even healthy people

Problem 1

The concentration of **PM_{2.5}** in ambient air is found to be **75 µg/m³**.

Using the **AQI breakpoint table**, calculate the **Air Quality Index (AQI)** for this pollutant and identify the **corresponding air quality category**.

Problem 1

Given:

$$\text{PM}_{2.5} = 75 \text{ µg/m}^3$$

Step 1: Identify range

75 lies in **61–90 µg/m³** → **AQI range 101–200**

$$B_{LO} = 61, B_{HI} = 90$$

$$I_{LO} = 101, I_{HI} = 200$$

Step 2: Apply formula

$$I = \frac{200 - 101}{90 - 61} (75 - 61) + 101$$

$$I = \frac{99}{29} \times 14 + 101$$

$$I = 47.79 + 101 = 148.79$$

Answer:

AQI ≈ **149 (Moderate)**

Problem 3

The measured concentration of **PM_{2.5}** in an urban area is **110 µg/m³**.
Calculate the **AQI** and classify the air quality based on standard categories.

Problem 2

Given:

$$\text{PM}_{2.5} = 110 \text{ µg/m}^3$$

Step 1: Range

$$91-120 \rightarrow \text{AQI } 201-300$$

$$B_{LO} = 91, B_{HI} = 120$$

$$I_{LO} = 201, I_{HI} = 300$$

Step 2: Calculation

$$I = \frac{300 - 201}{120 - 91} (110 - 91) + 201$$

$$I = \frac{99}{29} \times 19 + 201$$

$$I = 64.86 + 201 = 265.86$$

Answer:

AQI $\approx$ **266 (Poor)**

Problem 3

In an industrial region, the **PM_{2.5}** concentration is observed to be **160 µg/m³**.
Compute the Air Quality Index (AQI) and determine the level of pollution.

Problem 3

Given:

$$\text{PM}_{2.5} = 160 \text{ µg/m}^3$$

Step 1: Range

$$121-250 \rightarrow \text{AQI } 301-400$$

$$B_{LO} = 121, B_{HI} = 250$$

$$I_{LO} = 301, I_{HI} = 400$$

Step 2: Calculation

$$I = \frac{400 - 301}{250 - 121} (160 - 121) + 301$$

$$I = \frac{99}{129} \times 39 + 301$$

$$I = 29.9 + 301 = 330.9$$

Answer:

AQI $\approx$ **331 (Very Poor)**